

## Outline

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## Objectives

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- 1 To explain the importance of cost estimation for managerial decision-making.
- 2 To explain the different methods of cost estimation and their relative advantages and disadvantages.
- 3 To describe the different types of empirical study which are used in cost estimation.
- 4 To explain the types of problem which are encountered in statistical cost estimation.
- 5 To explain how these problems apply in different ways to short-run and long-run situations.
- 6 To explain how these problems can be overcome.
- 7 To describe and interpret the different types of cost function in mathematical terms.
- 8 To explain the specification and estimation of the learning curve.
- 9 To examine and interpret the findings of various empirical studies.

## 7.1 Introduction

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After discussing the theory of demand in Chapter 3 we proceeded to examine the estimation of demand functions in the following chapter. Now that we have discussed the theoretical aspects of production and cost we need to examine the estimation of cost functions.

### 7.1.1 Importance of cost estimation for decision-making

We have already seen in the previous chapter why managers need to understand the nature and application of cost functions in various aspects of decision-making. Since these functions are not self-evident, they have to be estimated using an appropriate method. Managers cannot apply their knowledge of the firm's cost functions unless these have been estimated in the first place. Therefore, cost estimation is important for the same reasons as an understanding of cost theory, that is:

- 1 To determine the pricing of the firm's products.
- 2 To determine the other components of the marketing mix.

- 3 To determine the optimal output of the firm in the short run and the relevant input mix.
- 4 To determine the optimal scale of operation of the firm in the long run.
- 5 To determine whether to accept or refuse an order from a potential customer at a particular price.
- 6 To determine the impact of potential mergers and acquisitions on unit costs.

### 7.1.2 Types of cost scenario

The word **scenario** is used here to indicate a distinction between a cost function's time period and a cost function's mathematical form. Scenario in this context refers to time period, although we shall see that there is some relationship between time period and mathematical form.

It is vital for an investigator to determine which type of cost scenario is desired before collecting the data which are necessary to estimate the relevant cost function. This is because different scenarios require different types of data in order to estimate them. Essentially there are three possible main scenarios involved in cost estimation:

- 1 The short run – discussed in section 7.2.
- 2 The long run – discussed in section 7.3.
- 3 The learning curve – discussed in section 7.4.

The differences between these scenarios have already been explained in the previous two chapters, but the implications in terms of estimation have still to be discussed.

### 7.1.3 Methodology

As with demand estimation, there are different methods of estimation that can be used. However, once again we find that statistical analysis tends to be the preferred method, with its advantages being essentially the same as for demand analysis. Most of the remainder of this chapter will examine various aspects of statistical cost estimation, but before we move on to this it is helpful to consider two alternative methods of estimation: *engineering analysis* and the *survivor method*.

#### *a. Engineering analysis*

This is mainly concerned with **estimating physical relationships between inputs and outputs in order to estimate a production function**. This in turn can be used to obtain the cost function of the firm, once we know the prices of the inputs, and assuming that the firm produces with economic efficiency. In the short run the production process will often determine how the inputs are combined. Technical factors frequently determine, for example, how many workers can use a particular machine, how many hours a machine can operate in a day,

how much raw material and what components are required to manufacture a finished product, how much fuel and power is required, and so on. Often relationships here may be linear, so that, for example, producing twice as much output requires twice as much material; sometimes, however, as with fuel requirements, the relationship may be non-linear; a machine operating twice as fast may require more than twice as much fuel.

Having established production relationships, the next step is to price these inputs. This may involve some of the problems mentioned in the previous chapter, in that accounting data may be historical rather than current. Also the data may be aggregated in such a way as to make it difficult to estimate prices for particular types of labour or materials. The final step with this approach is to derive the cost function on the basis that optimal production methods are used. This may also present problems, since there may be some flexibility in terms of input substitution; for example, taking the variable factors in the short run, one material might be substituted for another, such as steel for aluminium in car production. Thus the firm has to determine the least-cost cost function, in other words the cost function that involves producing each output at minimum cost. This is the familiar mathematical problem that we have seen in Chapter 5, where the objective is to minimize total cost subject to a series of output constraints (shown as shifting isoquants). The only difference is that in this case both inputs are variable in the short run.

There are three main shortcomings of engineering analysis.

1. Some production processes are highly complex, involving a large number of inputs. This makes it difficult to determine the mathematical model of the relationships between these inputs and output. The problem is even more difficult if there is more than one output and there are interdependencies between these outputs, for example if a plant is producing different models in a product line.

2. No consideration is given to indirect costs. Thus management, administrative and distribution inputs and costs are ignored in the analysis. Since these are also related to output in practice, they can have a considerable impact on the firm's cost function.

3. There is often no consideration of optimization of the input-output relationships. Existing combinations of inputs and outputs are often used in the analysis, with no regard as to whether these are optimal. In order to determine the optimal combinations the aspect of flexibility of input substitution discussed above must be examined; this can make the analysis considerably more difficult, and may require some costly, and risky, experimentation by the firm in terms of varying its production processes.

#### *b. Survivor method*

This method was originally developed by Stigler,<sup>1</sup> and only applies to long-run cost estimation. The method does not rely on the use of accounting data, and therefore avoids the problems arising from the unreliability of such data. The

method simply involves categorizing the firms in an industry according to size, observing the industry over a relatively long time period, and **recording the growth or decline of the different size categories**. Some conclusion is then formed regarding the most efficient size of firms in the industry according to which size categories are growing fastest. Stigler applied this technique to the steel industry in the United States, using data from 1930 to 1951, and observed that the medium-size firms (between 2.5 per cent and 25 per cent of industry capacity) had gained most in terms of total market share. He concluded that these firms must be the most efficient and that therefore the industry featured a U-shaped LAC curve. Apart from its avoidance of unreliable accounting data, the other advantage of the survivor method is its simplicity. It does however suffer from three serious problems.

1. It does not estimate costs or unit costs at any level of output, and therefore is not very useful for most aspects of managerial decision-making. One exception may be the consideration of mergers and acquisitions.

2. It assumes that the industry is highly competitive, so that firms that are performing below maximum efficiency will not survive. It thus ignores market power and barriers to entry, discussed in the next chapter.

3. It ignores changing technology and its impact on optimal size. In the steel industry for example it has become easier in recent decades for smaller firms to compete by using smaller plant size with new technology, resulting in unit costs comparable to larger firms. The survivor method may not reveal such a reduction in minimum efficient scale if it has only occurred in the more recent years of the long period under consideration.

### c. *Statistical analysis*

In view of the problems described above, statistical methods are often used to estimate cost functions. These overcome some of the problems, but are not without problems of their own, as we shall see. Much of the pioneering work in this area was carried out by Dean,<sup>2</sup> who also wrote the first textbook on managerial economics. He conducted some of the empirical studies that are discussed later in the chapter. Johnston reviewed and extended this work,<sup>3</sup> and much of the material in this chapter is based on their work.

The procedure for statistical cost estimation is essentially the same as for demand estimation. The same seven steps are involved:

- 1 *Statement of a hypothesis*. An example here might be that a firm's short-run total cost function is linear.
- 2 *Model specification*. This means determining what variables should be included in the cost function and what mathematical form or forms such a relationship should take. These issues are again determined on the basis of economic theory and prior empirical studies; obviously output is the most important explanatory variable. Various alternative models may be specified at this stage, since economic theory is often not robust enough to be definitive regarding the details of the form of model.

- 3 *Data collection.* This stage can only be performed after the cost function has been specified, otherwise it is not known for which variables we have to collect data.
- 4 *Estimation of parameters.* This means computing the values of the coefficients of the variables in the model. These effects can be measured in different ways, for example in terms of the marginal effects and elasticities already discussed. Clearly we must use some technique to estimate these values and the method of OLS regression will again be used in this context.
- 5 *Selecting the best model.* Once several alternative models have been estimated, it is necessary to examine how well each model fits the data and which model fits best. The situation is somewhat different here from the demand situation. In the latter case, as we have seen, economic theory is not usually robust enough for us to specify a mathematical form. However, in cost situations, theoretical considerations may be dominant in selecting suitable forms, as will be seen in the remainder of this chapter. If the fit is not good it may be necessary to return to step 2 and respecify the model before moving on to the next stage.
- 6 *Testing a hypothesis.* In the example above this is determined by comparing the goodness of fit, as described in the previous stage.
- 7 *Forecasting.* Once the appropriate cost function has been estimated cost forecasts for different outputs can be computed.

## 7.2 Short-run cost estimation

Short-run cost functions assume, as we have seen, that at least one factor is fixed. Thus changes in cost are caused mainly by changes in the level(s) of the variable factor inputs. This has important implications regarding the type of empirical study used. Different measures of cost are sometimes used as the dependent variable, depending on the availability of data; thus the cost function may be specified as:

$$VC = f(Q) \quad (7.1)$$

$$TC = a + f(Q) \quad (7.2)$$

$$AVC = f(Q)/Q \quad (7.3)$$

$$ATC = [a + f(Q)]/Q \quad (7.4)$$

$$MC = f'(Q) \quad (7.5)$$

In each case it is assumed that there is only one explanatory variable, output. Once one particular form of cost function has been estimated, other forms can be obtained mathematically from that form.

### 7.2.1 *Types of empirical study*

Various approaches are possible here. In some cases, experiments can be specifically designed for the purposes of cost estimation, or panel data may be used, as discussed in the chapter on demand estimation. However, as with demand studies, there are two main types of study that can be performed: time-series and cross-section. Each involves different constraints and problems. These are explained briefly here and then expanded in the next subsection.

#### *a. Time-series studies*

In this case the researcher needs data relating to a specific plant or firm over a period of time. However, it is important that the time period involved be relatively short, since in order to qualify as a short-run analysis neither the size of the plant, the scale of the firm nor the technology in use should change. There must be sufficient time periods, and there must also be some variation in production levels from one period to another, for the analysis to be reliable. Otherwise the estimated coefficients of the variables in the equation will have large standard errors. To obtain sufficient observations in the short time period such observations may have to be made at monthly intervals, or even more frequently. This may pose problems, which are discussed later.

#### *b. Cross-section studies*

A researcher can estimate a short-run cost function for an industry by examining different firms of the same scale. However, a firm can only use a cross-section study to estimate its particular cost function if it has several plants. Since managers are usually concerned with the analysis of firm-specific data, this places a constraint on the use of cross-section studies. If the firm does operate several plants, these plants must be of the same size, and use the same technology, in order for the study to qualify as a short-run analysis. Furthermore, these plants must all be producing the same product, and involve the same stage of the production process, in other words the firm must be horizontally integrated. Again, for the results to be reliable, there must be some variation between the outputs of the different plants. Given the constraints described above, time-series studies are more commonly used for short-run cost estimation.

### 7.2.2 *Problems in short-run cost estimation*

In Chapter 4 on demand estimation it was seen that there were a number of problems related to the estimation process. There are again problems with cost estimation, but they tend to be of a different nature. The main problematic area is in data collection and measurement, as explained below.

#### *a. Dynamic environment*

If time-series analysis is used it must be recognized that the cost function is really a dynamic relationship that is changing over time. Thus there is an

identification problem similar to the situation discussed under demand estimation. This means that each observation could relate to a different cost function if care is not taken to keep other factors equal; if other factors cannot be kept equal then certain adjustments must be made to the values of the variables recorded. One of the most important factors that needs to be kept equal is the product quality; this is more difficult to achieve with a time-series study, as quality improvements may be being made unknown to the researcher. This problem of a dynamic environment is explained in more detail in conjunction with the next problem.

### *b. Use of accounting data*

Researchers are inevitably constrained to using data collected and recorded by the firm's accountants. As already seen in the last chapter, this information has been collected and recorded for different purposes from those of the researcher: to conform to legal requirements and externally imposed accounting standards, and for other external uses like providing shareholder information. Managers who want to use the data for internal purposes, therefore, are faced with a number of problems. The principles involved are the same as those discussed in the previous chapter: the relevant costs for decision-making purposes are economic costs rather than accounting costs, current costs rather than historical costs, and incremental rather than sunk costs. The application of these principles to cost estimation involves the following specific aspects.

**1. Adjustment for changes in prices.** Prices of labour, materials and other inputs must be adjusted so that current prices are used. The best way of doing this is to use a specific cost index for the relevant input. Say, for example, that labour costs were recorded as being £10,000 for a particular time period, and that wage rates had increased by 12 per cent since then; these labour costs should be recorded at the estimated current costs of £11,200. If a specific cost index is not available, a general price index like the wholesale price index can be used (this is preferable to the retail price index, as being more relevant to input prices).

**2. Measurement of depreciation.** Accountants often measure depreciation on an essentially arbitrary basis from an economist's viewpoint, for example to perform easier calculations (straight-line method), or to reduce tax liability. This means that depreciation is often calculated on a time-related basis, whereas economists are interested in the usage-related aspect of depreciation. Furthermore, accountants calculate depreciation based on historical cost, whereas economic depreciation should be based on replacement value.

### *c. Multiproduct firms*

We have assumed up to this point that the firm is producing a single product. If a firm produces different products from different plants cost estimation may



not be too difficult; different cost functions can be estimated for each product from each plant. There is still a problem of overhead allocation, discussed below, but the situation is still more simple than if more than one product is produced from a single plant, as is often the case with a product line. There are two possible approaches to this problem.

**1. Combination of products into a single output variable.** This is most easily explained by means of an example. Say that a firm produces two models of car, model A and model B. In a particular time period a plant may produce 2,000 of A and 3,000 of B. Since the models may have different prices according to the different production costs involved, it may be preferable to take these prices into account in measuring total output, rather than just adding the crude output figures together. Thus if A is priced at £8,000 and B is priced at £12,000, the total value of output is £52 million for the period. This can then be divided by an average price of £10,000 to obtain a proxy or combined measure of output of 5,200 units. This figure can then be compared with outputs from other periods, provided that once again allowance is made for any changes in prices, this time of the outputs.

**2. Estimating separate cost functions for different products.** The main problem here is the allocation of fixed overhead costs. Accountants again tend to allocate these on an arbitrary basis, sometimes on the basis of the **net realizable value** of the different products. Net realizable value is usually defined as **revenues minus direct costs**. This basis for allocating overheads causes problems for the estimation of cost functions because it treats a cost category as being variable when it is really fixed. The result is that this method can result in nonsensical cost functions with negative average variable costs, as shown by Dobbs.<sup>4</sup> The problem of estimating cost functions becomes even more serious when there are cost interdependencies among the different products.

#### *d. Timing of costs*

It has already been stated that in a time-series approach to short-run cost estimation the total time period for the study must be of relatively short duration, and this can necessitate the use of short time intervals, such as a month or less, for observations in order to obtain sufficient observations. The lower time limit for these intervals is set by the accounting periods for the firm. Costs, or at least some costs, may not be recorded on a weekly basis.

However, there is another problem related to short observation intervals, and this concerns **spillover effects**. There is a danger of **recording costs in the wrong time period**; for example, billing dates may not match production and usage. Some costs may appear before or after the period to which they relate. In addition to this, some costs can be scheduled according to convenience rather than production; this applies particularly to maintenance, which is often scheduled for quiet periods when production is low. In reality this maintenance cost is related to previous periods when production levels were higher. Only

when researchers are very knowledgeable regarding production activity and the accounting practices of the firm are they likely to be able to avoid these pitfalls.

In addition to the problems described above, it should also be noted that the function being estimated is based on actual costs that have been incurred by the firm(s) in the study; if the firm or firms have not been operating efficiently the cost function estimated will not be an optimal cost function. It will tend to overstate the 'true' values of the cost parameters.

### 7.2.3 Different forms of cost function, interpretation and selection

#### a. Forms

We saw in the previous chapter that cost functions can take a variety of forms in the short run. The polynomial form is the most common form, consisting of linear, quadratic and cubic functions. In each case the relationships between total cost, average cost and marginal cost were examined. Therefore, as stated earlier in this chapter, it does not matter which of these is specified in the cost model, since the other measures can be calculated mathematically from the measure specified. This is briefly reviewed below.

The marginal cost and average variable cost relationships have been seen to have a U shape in many situations, because of increasing and then diminishing returns. This translates mathematically into a quadratic unit cost function of the general form:

$$MC = b + cQ + dQ^2 \quad (7.6)$$

We can then obtain the TC function by integration, producing the following cubic:

$$TC = a + bQ + (c/2)Q^2 + (d/3)Q^3 \quad (7.7)$$

In this case  $FC = a$  and  $VC = bQ + (c/2)Q^2 + (d/3)Q^3$

Thus

$$AFC = a/Q \quad \text{and} \quad AVC = b + (c/2)Q + (d/3)Q^2 \quad (7.8)$$

and

$$ATC = a/Q + b + (c/2)Q + (d/3)Q^2 \quad (7.9)$$

Alternatively, a particular form of TC function may be specified; in this case the MC function can be derived by differentiation. It can be easily seen that if the TC function is the quadratic:

$$TC = a + bQ + cQ^2 \quad (7.10)$$

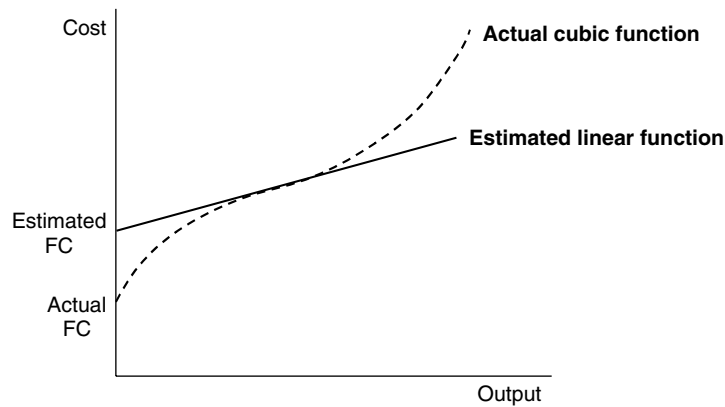


Figure 7.1. Error from mis-specifying the cost function.

then the MC function is linear:

$$MC = b + 2cQ \quad (7.11)$$

and AVC is also linear, but with half the slope:

$$AVC = b + cQ \quad (7.12)$$

An example involving this type of analysis was given as solved problem [SP6.3](#).

The simplest type of cost function is the linear form:

$$C = a + bQ \quad (7.13)$$

where FC is given by  $a$ , and MC and AVC are both given by  $b$ .

#### *b. Interpretation*

Although regression analysis can estimate the values of these coefficients, care must be taken in their interpretation, particularly of the value of  $a$ . This is because the function estimated is only valid over the output range of the data; this is unlikely to include zero output. Thus the interpretation of the value of  $a$  as being the level of fixed costs often involves a huge extrapolation. If the function estimated is not valid over other parts of the output range the result could be a large error in the estimation of fixed costs. In essence this is a mis-specification error. This is illustrated in Figure 7.1, where a cubic function is mis-specified as linear.

#### *c. Selection*

Since the researcher cannot normally be confident regarding which mathematical form is the best one to use from a theoretical viewpoint, the procedure for selecting the best model is essentially the same as for estimating demand functions: different mathematical forms are estimated, usually linear, quadratic and cubic, and the one which fits best (on the basis of adjusted  $R^2$ ) is selected.

### 7.2.4 Implications of empirical studies

Many empirical studies of short-run cost functions have been conducted, both at the level of the individual firm and at industry level, since the 1930s. The landmark studies, which are frequently quoted at length in textbooks, are those by Dean<sup>5</sup> in 1941 (hosiery) and Johnston in 1960<sup>6</sup> (road passenger transport). Both these researchers also estimated cost functions in other industries, as have other researchers, and there are now extensive studies of the furniture, food processing, steel, electricity, coal, cement and other manufacturing industries. It is not necessary to consider any of these in detail here, since they come to largely the same conclusions. Walters<sup>7</sup> has summarized many of the earlier findings, concluding that they generally do not support the hypothesis of the U-shaped average cost curve. More recent studies also tend to indicate that total cost functions are frequently linear, with constant marginal costs, at least in the normal range of output of firms. For example, in a study of a firm producing plastic containers,<sup>8</sup> it was found that for all ten products manufactured by the plant the linear cost function provided the best fit.

Economists have come to some different conclusions based on these findings, in order to reconcile them with the U-shaped average cost curve of economic theory:

- 1 The data used have related only to a limited range of output, the normal output range of the firms studied. As already explained in the previous chapter, it may well be that the law of diminishing returns only starts to take effect as firms approach full capacity, perhaps at around 95 per cent of maximum output. Firms do not normally operate at this level, and indeed managers try to avoid such operation because of the rising unit costs.
- 2 The studies have actually been long-run rather than short-run, because capital inputs have varied, allowing the ratio of labour to capital to remain constant.

Therefore, it is not necessary to re-evaluate the premises of economic theory in view of the empirical findings. Furthermore, there is one other finding in practice that tends to confirm that, ultimately, marginal cost curves must be rising at higher levels of output. This is the fact that an increase in demand for a product results in an increase in both the quantity supplied and the price of the product in the short run. It will be seen in the next chapter that an upward-sloping supply curve is dependent on an upward-sloping marginal cost curve.

## 7.3 Long-run cost estimation

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Many of the basic aspects of cost estimation in the long run are similar to those for the short run, thus the structure of this section is the same as for the previous one. However, we shall see that there are some different types of constraints and problems associated with the long run, that different mathematical forms are

often used, and that the implications of empirical studies are of a different nature.

### 7.3.1 *Types of empirical study*

As with short-run studies, empirical studies can be classified into time-series and cross-section categories. These are now discussed separately, in terms of the constraints and procedures involved.

#### *a. Time-series studies*

In this case it is necessary to use a time period that allows the firm or plant to experience several changes in capacity; this is because the aim of such a study is to estimate the effect on costs of these changes in capacity in terms of economies and diseconomies of scale. We shall see that this creates some serious problems for this type of study.

#### *b. Cross-section studies*

These involve collecting data from firms or plants of different sizes or scales. If a firm-specific function is desired then this is only possible if the firm has several plants, each producing the same product, involving the same stage of the production process and using the same technology, but with the plants being of different sizes. This constraint may make such a study impracticable, since it may be that the long-run average cost curve is U-shaped, resulting in only a certain plant size being efficient. In this situation it would not be sensible for a firm to operate different plants with different capacities. However, if a cross-section study is possible, it is generally preferable to a time-series study, because the problems involved tend to be less serious. These problems are now examined.

### 7.3.2 *Problems in long-run cost estimation*

Since the nature of these problems is different according to whether the study is time-series or cross-sectional, it is again best to discuss these problems separately.

#### *a. Time-series studies*

The main problem here is that, in order to allow the firm sufficient time to change its capacity several times, such a long time period is involved that technology also changes. It then becomes difficult to separate the effects of increased scale from the effects of improved technology. The problem can sometimes be overcome by the use of a dummy variable (or more than one such variable) in the cost function to represent different levels of technology.

Another problem is that the firm may not be operating at maximum efficiency with each plant size. It is always assumed in drawing the long-run average cost curve as an envelope to the short-run curves that maximum

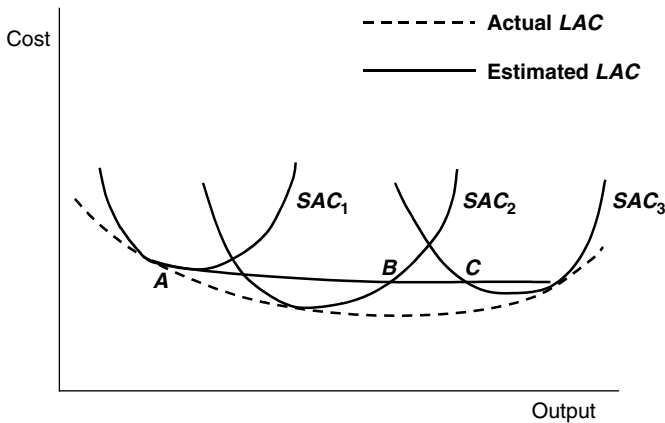


Figure 7.2. Estimation error arising from sub-optimal operation of plants.

efficiency is achieved from each scale of operation. This problem may cause the LAC curve to be estimated wrongly, as shown in Figure 7.2. In this case the firm moves from point A to point B to point C and it appears that the LAC curve shows no economies or diseconomies of scale.

Other problems discussed in short-run cost estimation are also often present, for example allowing for changes in input prices, the correct measurement of depreciation and overhead costs, and the measurement of output for multiproduct firms. The problem of changes in input prices becomes greater in the long run than in the short run. This is because a greater variation is often present and the longer time period allows managers to adjust input combinations to these different prices. If prices are all adjusted to a particular base, this ignores these managerial adjustments and prevents an accurate analysis of the cost-output relationship. On the other hand, the problem of the timing of costs and the matching of costs to production periods is usually less serious, since the intervals between observations tend to be longer.

#### *b. Cross-section studies*

These studies often avoid some of the more difficult problems above. For example, the problem of changing technology may not occur if observations are all made at the same time period. The problem of allowing for changes in input prices may also not be relevant for the same reason. However, these problems may not be altogether avoided, for the following reasons:

- 1 Different plants may use different technologies, especially if they have been built at different times. As with time-series studies, it may be possible to allow for this by using dummy variables.
- 2 Different plants may face different input costs, especially if they are located in different geographical areas. Labour, raw materials or power and fuel prices may vary from area to area; transportation costs may also be different. It may be possible to allow for these variations by some adjustment to a

constant base, but this creates the further problem of ignoring managerial adjustments, described above.

- 3 Plants may not operate at maximum efficiency, leading to the same problem as that outlined for time-series studies above, and illustrated in Figure 7.2.
- 4 Different firms may record accounting data differently, if an industry study is being conducted. This applies especially to the measurement of depreciation, the valuation of inventory and the allocation of overhead expenses. Different firms may pay their labour differently also; this applies particularly to managers, and most recently to managers in high-tech companies who are often paid largely in terms of stock in the company.

### 7.3.3 *Different forms of cost function*

Whereas polynomial functions tend to be the favoured form for short-run cost functions, it has been seen in the previous chapter that, in the long run, power functions are the favoured form. These can be expressed as follows (assuming a single product firm):

$$C = aQ^b \quad (7.14)$$

In this case we are using the symbol  $C$  for total costs, since in the long run there is no distinction between total costs and variable costs; all costs are variable. Alternatively, the cost function can be written in log-linear form:

$$\ln C = a' + b \ln Q \quad (7.15)$$

If a dummy variable is required to indicate a different level of technology this equation can be expanded to:

$$\ln C = a' + b \ln Q + cT \quad (7.16)$$

where  $T=0$  for the base technology and  $T=1$  for a newer technology. Obviously additional dummy variables are necessary if there are more than two different levels of technology.

The advantage of this power form, as we have again seen in the previous chapter, is that it allows us to estimate cost elasticities, which in turn are an indicator of the existence of economies or diseconomies of scale. The solved problem SP7.2 involves this type of analysis, including the problem of allowing for different technologies.

### 7.3.4 *Implications of empirical studies*

Again, much of the early empirical work involving long-run studies was conducted by Dean and Johnston. Both researchers found U-shaped average cost curves, in the shoe retailing<sup>9</sup> and building society<sup>10</sup> industries respectively, although in the first case the rising average costs were not interpreted as due to

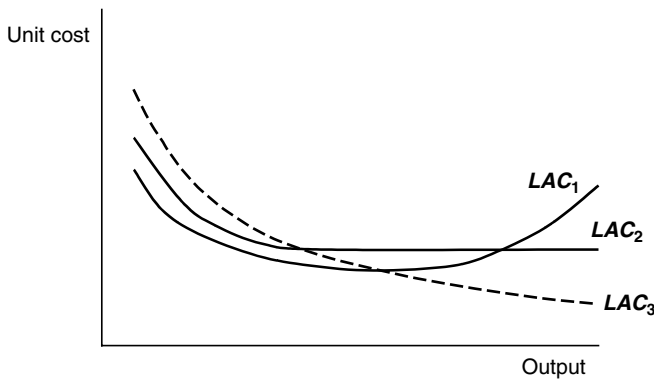


Figure 7.3. Different shapes of LAC curve.

diseconomies of scale. In other studies by Johnston, of electricity, road passenger transport and life assurance, he found declining average costs.

More recent empirical work has examined cost structures in a wide variety of manufacturing industries, and there have also been studies relating to service industries such as banking and information technology. Industries can generally be categorized into three classes:

- 1 Those with U-shaped LAC curves – specialized retailing and many services.
- 2 Those with L-shaped LAC curves – general and chain retailers; most manufacturing.
- 3 Those with continuously falling LAC curves – mass manufacturing and public utilities.

These different situations are shown graphically in Figure 7.3.

The extent of economies of scale in a firm or industry is often measured in terms of the **cost gradient**. This measures **the increase in unit cost as a result of a firm or industry operating at a specified percentage of the optimal size, usually 50 per cent**. An extensive summary of studies concerning economies of scale in manufacturing in the UK, the USA and Germany has been carried out by Pratten.<sup>11</sup> This showed cost gradients at 50 per cent of minimum efficient scale varying from 1 per cent (in footwear) to 36 per cent (in book printing).

Another method of measuring the extent of economies of scale is to compute the minimum efficient scale (MES) as a percentage of total output in a national or international market. Certain industries had a higher MES than the size of the UK market, for example dyes, aircraft and diesel engines. In addition, many firms featured economies of scale of a non-technical type at the level of the firm rather than at the level of the plant, with these economies continuing to occur at much higher levels of output. Pharmaceuticals and cars are good examples. In the latter case the optimal scale from a technical viewpoint has been found to be up to 2 million cars per year,<sup>12</sup> but non-technical economies relating to finance and research and development increase this optimal output to 5 million cars per year.



Various empirical studies have also measured economies of scope. Apart from the measure described in the previous chapter, another measure sometimes used is the percentage increase in average costs that results from producing only half of the number of models or product forms. With motor vehicle production this appears to be higher than other industries, at around 8 per cent; with pharmaceuticals, electrical machinery, office machinery and domestic appliances the increase has been estimated at 5 per cent.<sup>13</sup>

What are the main implications for managerial decision-making arising from these studies? Some of the most important are as follows:

- 1 The optimal size of the firm depends on the type of industry in which it is operating.
- 2 In some industries like pharmaceuticals and car manufacturing, where the  $LAC$  curve resembles  $LAC_3$  in Figure 7.3, bigger is better from the point of view of cost efficiency. This encourages mergers and acquisitions, and also joint ventures. These are often on an international scale; in silicon chip manufacturing, for example, Siemens and IBM, Toshiba and Motorola, NEC and AT&T, all have some kind of agreement between them. In car manufacturing there have been various international mergers, such as Daimler-Chrysler and Renault-Nissan; joint venture agreements are also becoming more common, as firms like Mitsubishi and Chrysler have co-operated on research and using common platforms.
- 3 In those industries where the **MES** is greater than the size of the national market it is most efficient for production to be undertaken by a multinational firm, in order to gain the full benefit from the economies of scale available.

There now follow three case studies involving banking, airlines and electricity generation, all of which have been studied extensively in empirical terms. These will allow the reader to see how different cost considerations apply in different types of industry.

### Case study 7.1: Banking

There are various theoretical reasons why economies of scale should occur in the banking industry:

- 1 Specialization of labour. There is considerable scope for this as cashiers, loan officers, account managers, foreign exchange managers, investment analysts and programmers can all increase their productivity with increased volume of output.
- 2 Indivisibilities. Banks make use of much computer and telecommunications technology. Larger institutions are able to use better equipment and spread fixed costs more easily.
- 3 Marketing. Much of this involves fixed costs, in terms of reaching a given size of market; large institutions can again spread these costs more easily.
- 4 Financial. Banks have to raise finance, mainly from depositors. Larger banks can do this more easily and at lower cost, meaning that they can afford to offer their depositors lower interest rates.

There are also reasons why banks should gain from economies of scope; many of their products are related and banks have increasingly tried to cross-sell them. Examples are different types of customer

account, accounts and credit cards, accounts and mortgages or consumer loans, and even banking services and insurance.

There has also been a spate of bank mergers and acquisitions in recent years, often involving related institutions like building societies, investment banks and insurance companies. Many of these institutions have been very large in size, with assets in excess of \$100 billion. Examples are Citibank and Travellers Insurance (now Citigroup), Bank of America and NationsBank, Chase Manhattan and J. P. Morgan, HSBC and Midland; both NatWest Bank and Abbey National Bank in the UK have been the object of recent takeover bids. This would tend to support the hypothesis that 'bigger is better'.

The empirical evidence, however, is not supportive of the 'bigger is better' policy that many banks seem to be following. A number of empirical studies have been carried out regarding commercial banking and related activities, in both Europe and the United States. Some US studies in the early 1980s found diseconomies for banks larger than \$25 million<sup>14</sup> or \$50 million<sup>15</sup> in assets, a very small size compared with the current norm (the largest banks now have assets in excess of \$500 billion). More recently a greater availability of data has enabled research to be carried out on much bigger banks, as deregulation in 1980 led to interstate banking in the United States.

Shaffer and David<sup>16</sup> examined economies of scale in 'superscale' banks, that is banks with assets ranging from \$2.5 billion to \$120 billion in 1984. They estimated that the minimum efficient scale of these banks was between \$15 billion and \$37 billion in assets, and that these larger banks enjoyed lower average costs than smaller banks.

Many of the studies have been summarized by Clark in the USA.<sup>17</sup> In particular, Clark's conclusions were that there are only significant economies of scale at low levels of output (less than \$100 million in deposits). Furthermore, it appeared that economies of scope were limited to certain specific product categories, for example consumer loans and mortgages, rather than being generally applicable.

### Questions

- 1 What shape of long-run average cost curve appears to be appropriate for the commercial banking industry?
- 2 What mathematical form of cost function would be most appropriate to use to test the existence of economies of scale in banking?
- 3 What factors might cause the *LAC* curve to flatten out at high levels of output?
- 4 In view of the empirical evidence, what factors do you think might be responsible for the current trends of increasing size and mergers?

## 7.4 The learning curve

The concept of the learning curve was introduced in the previous chapter; it was seen that there are various factors involved which cause individuals and firms to learn better ways of doing something with experience. We now need to consider how to estimate a learning curve in practice.

### 7.4.1 Types of specification

The general model of the learning curve has been expressed as a power function:

$$U = aT^b \quad (7.17)$$

which is a repetition of (6.18). However, within this general model, there are a number of variations in terms of how the variables can be defined and measured. The *T* variable almost always represents cumulative output, although sometimes it is defined as a time variable, showing how unit costs change over time. There are many possible ways of defining *U*. It most often relates to a

### Case study 7.2: Airlines

The airline industry has very marked differences in structure between the United States and Europe. US airlines were deregulated in 1978, leading to intense competition on many routes, bankruptcies and mergers. In Europe the airlines have remained highly regulated, with governments protecting their own largely state-owned airlines. The deregulation in the US has led to much lower fares per passenger-mile and has caused some radical changes in structuring and therefore unit costs.

Economies of scale arise if airlines can make use of a single type of plane. For example, Southwest Airlines only flies the Boeing 737, while the struggling US Airways uses a number of different aircraft. This was largely because US Airways came into existence as a result of a series of mergers. Southwest can therefore reduce its maintenance costs, crew training and scheduling costs compared with US Airways. This was reflected in the fact that in 1993 its costs were only 7 cents per available seat mile, compared with over 11 cents for US Airways.

Economies of scope are also very important in the airline industry. Clearly it is cheaper for an airline to carry passengers and freight in the same aircraft than carry them in separate aircraft. However, the most important economy of scope arose from a change in routing: instead of having a large number of point-to-point services between different cities, airlines like Southwest and Continental switched to using a 'hub and spoke' system. This meant that customers flew from one city to a central hub, and then from there to their final destination, as shown in Figure 7.4. This gave these

airlines a number of advantages. They could fly a much larger number of routes, in terms of joining city-pairs, with a given number of aircraft and miles flown. This would attract more passengers. Since these passengers all landed at the hub, before being transferred to their final destination, it allowed the airlines to fill the flights more comprehensively. It also became possible to combine passengers with different characteristics more effectively; thus routes that largely carried business passengers could now attract vacation customers.<sup>18</sup>

The above economies of scale and scope have not prevented many airlines from getting into serious financial difficulties recently. Although the events of 11 September 2001 did not help the industry, there were already many problems brewing in terms of overcapacity and industrial relations. Many airlines now appear likely to go out of business unless they receive government subsidies, a controversial issue discussed in more detail in Chapter 12.

#### Questions

- 1 Explain how you would conduct an empirical study to investigate the availability of economies of scale in the airline industry.
- 2 What implications are there regarding mergers of different airlines?
- 3 What disadvantages might be incurred with the hub and spoke system compared with the point-to-point system?
- 4 Explain why the hub and spoke system relates to economies of scope rather than economies of scale.

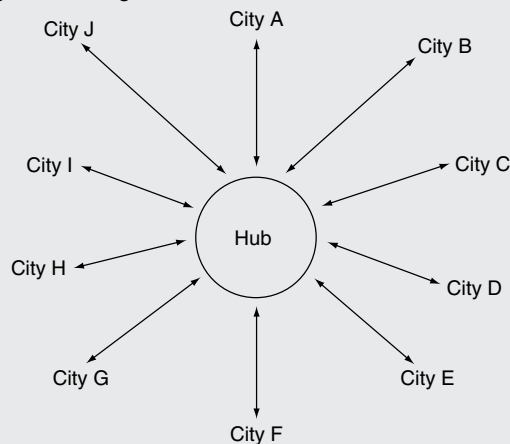


Figure 7.4. Hub and spoke airline connections.

### Case study 7.3: Electricity generation

This industry has been the subject of many empirical studies in different countries, going back to the 1950s. These studies have been facilitated by three factors:

- 1 Output is easy to measure (electricity produced, usually in billions of kilowatt-hours).
- 2 Different companies, or state utilities, have plants of different sizes.
- 3 Large amounts of data are available, since producers often have to record this for regulatory purposes.

Johnston<sup>19</sup> estimated long-run cost functions using both time-series and cross-sectional data, for the period 1928–47 and for twenty-three firms. Although a cubic total cost function was specified, he found that the cubic term was not statistically significant in any of the regression equations; the quadratic term was found to be significant in only six of the equations, and was positive in three cases and negative in the other three cases. In general, the linear total cost model tended to fit best.

Two major studies were conducted in the United States in the 1970s. Christensen and Greene<sup>20</sup> used a cross-sectional study involving 1970 data for 114 firms, specifying a power model to test for the presence of economies and diseconomies of scale. They found significant economies of scale below

about 20 billion kwh in output, relating to 85 per cent of the firms and nearly 49 per cent of total output. Between this output and 67 billion kwh they observed no significant economies of scale; this related to another 14 per cent of the firms and 45 per cent of output. Above this output they found that diseconomies of scale occurred for the one remaining firm, producing about 7 per cent of total output. This pattern of average costs is shown in Figure 7.5.

Another study by Huettner and Landon<sup>21</sup> examined seventy-four utility firms, using 1971 data, and specifying a cubic model. Their conclusions were broadly the same as for the previous study: the LAC curve was U-shaped.

Since the 1970s there have been great changes in the electricity generation industry. Alternative sources of power have become more important, owing to a combination of market forces, environmental concern and improvements in technology. The fossil fuels, coal and oil, have declined in importance, while natural gas, a much cleaner fuel, has gained in importance. Unfortunately, for many utility companies who have had to buy electricity recently from gas-powered sources, demand has far outstripped supply, and prices of gas have soared, in turn causing electricity prices on the wholesale market to soar. In northern

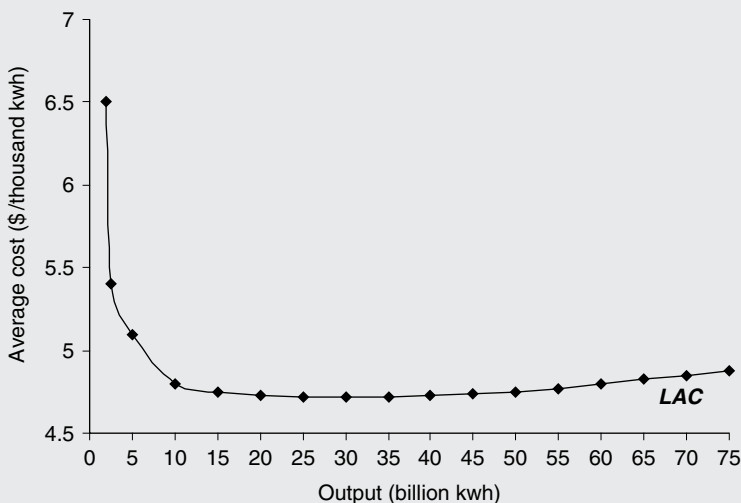


Figure 7.5. LAC curve for electricity generation.

California, for example, electricity prices are now ten times their level at the beginning of 2000.

Hydroelectric power, from dams, has also become more popular on a global basis, though this too has caused environmental and political problems.

When the electricity utility firms were deregulated in the United States from the 1980s, deregulation posed problems for many of them. The traditional means of producing electricity, whether coal- or oil-fired power stations or hydroelectric plant, involved very high fixed costs. Being sunk costs, these costs then became 'stranded' after deregulation, since electricity buyers could then buy from out-of-state suppliers; this practice is known as 'freewheeling'. Many utility firms lost customers to these competitive suppliers, which forced them to move to the left and upwards along their *LAC* curves, thereby increasing unit costs; this in turn made them even less competitive. These and other aspects of deregulation will be considered in another case study in Chapter 12.

Improvements in technology have led to cleaner methods of producing electrical power in recent years. Although still insignificant in terms of global market share, wind power is becoming more important, particularly in countries like Denmark and

Germany. The cost structure of this technology is also different, and many of the indivisibilities of the older methods of production are avoided. A large 1.65 megawatt turbine now costs less than £1 million to build; although this is 20 times that of the 1980s version, it produces 120 times as much energy. Given the problems with other sources of energy, in terms of environmental pollution or volatile costs, wind power may prove to be a major source of energy in the future, along with other new-technology sources like hydrogen power.

### Questions

- 1 Interpret the results obtained by Johnston in terms of the quadratic terms of three equations being positive and the quadratic terms of the other three being negative.
- 2 Explain the relative advantages and disadvantages of using power or cubic cost functions to estimate *LAC* curves for electricity production.
- 3 Explain the meaning of 'stranded costs'; why are they a problem?
- 4 What is happening to unit costs for producing wind power?

unit cost measure, although sometimes this is only one element of cost, like labour cost. *U* can also be defined as the level of input involved in producing a unit of output, or the amount of time taken to produce a unit of output.

Further variations of the model relate to how these variables are measured: cost, for example, is most often measured as the marginal cost of the last unit produced, but it can also be measured as average cost for the current production period, or as cumulative average cost since production began. The example in Table 7.1 relates to Nuprod plc and illustrates the different possibilities. In practice it is often easier to specify the model with *U* being the cumulative average cost (*CAC*), since this makes it easier to compute other values and make various types of forecasts. (*CAC*) is computed as cumulative total cost divided by cumulative output (*CTC/CQ*).

Before moving on to the application of the learning curve, it is necessary to consider one further aspect of specification. Sometimes it is desired to test whether a reduction in unit cost over time is caused by the learning curve or by economies of scale; this can be done by specifying a model that includes both factors. If we call output *Q* and cumulative output *X*, with average cost or marginal cost being *U*, the model becomes:

$$U = aQ^bX^c \quad (7.18)$$

Table 7.1. Nuprod plc: learning curve estimation

| Time period | Output, $Q$ (units) | Total costs, $TC(\pounds)$ | Cumulative total costs, $CTC(\pounds)$ | Cumulative output, $CQ (= \sum X)$ (units) | Cumulative average, costs $CAC (= Y)$ ( $\pounds$ ) |
|-------------|---------------------|----------------------------|----------------------------------------|--------------------------------------------|-----------------------------------------------------|
| January     | 23                  | 184                        | 184                                    | 23                                         | 8.00                                                |
| February    | 28                  | 122                        | 306                                    | 51                                         | 6.00                                                |
| March       | 29                  | 126                        | 432                                    | 80                                         | 5.40                                                |
| April       | 35                  | 128                        | 560                                    | 115                                        | 4.87                                                |
| May         | 30                  | 115                        | 667                                    | 145                                        | 4.60                                                |
| June        | 45                  | 107                        | 798                                    | 190                                        | 4.20                                                |

Alternatively the model can be specified in logarithmic form. In either case it separates the learning effect from the effect of economies of scale.

### 7.4.2 Application of the learning curve

We shall use the solved problem of Nuprod plc to illustrate the application of the learning curve. There is obviously only a small sample of observations in this example, with a greater chance of sampling error occurring, but the simplicity of the situation aids presentation. The main application of the learning curve, as with other aspects of estimation, is to make forecasts.

#### SP7.1 Learning curve estimation

Nuprod plc first started producing a particular product in January. It records output and costs over the next six months as shown in the accompanying table. Nuprod's management are planning to produce fifty units in July.

| Time period | Output, $Q$ (units) | Total costs, $T(\pounds)$ |
|-------------|---------------------|---------------------------|
| January     | 23                  | 184                       |
| February    | 28                  | 122                       |
| March       | 29                  | 126                       |
| April       | 35                  | 128                       |
| May         | 30                  | 115                       |
| June        | 45                  | 107                       |

- Estimate a learning curve for the data.
- How well do the data fit a learning curve?
- Estimate the cumulative average costs for July.
- Estimate the cumulative total costs for July.
- Estimate the total costs for July.
- Estimate the average costs for July.
- Estimate the marginal cost for the last unit produced in July.

*Solution*

The first step is to transform the original data relating to output and costs into a form that can be used to estimate the relationship. We can now compute the learning curve equation from the last two columns, using regression analysis of a power form:

$$a. \quad Y = 19.91X^{-0.2970} \quad (7.19)$$

where Y is the cumulative average cost in £, and X represents cumulative output in units.

- b. It is now necessary to check the goodness of fit by examining the  $R^2$ . In this case this is over 99 per cent, showing very strong evidence of a learning curve. As seen in the previous chapter, the **learning rate** can be calculated as  $2^b$ . In this case the learning rate is 81 per cent. The interpretation of this is that **every time cumulative output doubles, the cumulative average cost becomes 81 per cent of what it was previously.**

- c. CAC for July:  $CQ = 190 + 50 = 240$   
 $CAC = 19.91(240)^{-0.2970} = \text{£}3.91$
- d. CTC for July:  $CTC = 3.91 \times 240 = \text{£}938$
- e. TC for July:  $TC = 938 - 798 = \text{£}140$
- f. AC for July:  $AC = 140/50 = \text{£}2.80$
- g. MC for last unit in July:  
 $\frac{d(CTC)}{d(CQ)} = a(b+1)X^b = 19.91(0.2970 + 1)240^{-0.2970} = \text{£}2.75$

### 7.4.3 Implications of empirical studies

As stated in the last chapter, the learning curve was first applied empirically in the 1930s to aircraft production. Since then it has been applied in many industries, with the fastest learning rates being found in the production of electrical components and microcomputing. In both cases the learning rate was estimated at 70 per cent. For various other industries, for example ball-bearings, plastics, equipment maintenance and life assurance, the learning rate has been estimated at below 80 per cent,<sup>22</sup> meaning that cumulative average costs are reduced by more than 20 per cent with a doubling of cumulative output. In oil refining on the other hand the learning rate has been estimated at only 90 per cent.

In general the learning rate appears to be faster under the following conditions:

- 1 Intensive use of skilled labour.
- 2 A high degree of capital intensity and research and development intensity.<sup>23</sup>
- 3 Fast market growth; this compresses the time involved under which learning takes place.

- 4 Supply and demand are highly interdependent, creating a virtuous circle; thus the learning effect reduces costs and price, and this in turn leads to a higher quantity demanded, more output and more learning.

## 7.5 A problem-solving approach

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There are many similarities here between cost estimation and demand estimation in terms of general procedure. Essentially the main stages are the same as in estimating demand:

- 1 *Stating a theory or hypothesis.* This must relate in some way to a cost relationship. An example of this is that there are economies of scale in a firm or industry.
- 2 *Model specification.* This involves identifying variables that may be relevant in the cost relationship, determining the relevant definitions, and considering the functional form. This will depend to some extent on whether the study is of a short-run, long-run or learning curve type.
- 3 *Data collection.* This involves determining whether to use time-series or cross-sectional data, and dealing with problems of definition, measurement and timing.
- 4 *Data processing.* This stage involves estimating the parameters of the model(s) and is largely mechanical, involving calculators and computers with the relevant programs.
- 5 *Interpretation of results.* This combines checking goodness of fit with statistical inference.
- 6 *Forecasting.* This involves substituting relevant values into the estimated equation, and clarifying any assumptions that are involved.

With most problems that students encounter, the first and third aspects are not relevant, because they are given a data set to work with, although sometimes some processing of these data is necessary before estimating the relevant relationship. This is particularly important with the learning curve. The most fundamental problem is model specification. With most problems that students face, a linear or polynomial function is used for short-run situations and a power function is used for long-run situations. As with demand estimation, a lack of understanding of the theoretical underpinnings of the regression model can also cause problems; this is particularly relevant in interpreting the significance of the regression coefficients.

In practice the third stage of the process, data collection, may pose considerable problems to researchers, and these problems have been discussed in some detail in the chapter to try to give students some flavour of this aspect.

The following solved problems are intended to acquaint the student with how to perform the various stages outlined above. Since there are three main



cost scenarios, short-run, long-run and learning curve, and an example has already been given of the application of the learning curve, only two further solved problems need to be presented.

### SP7.2 Short-run cost estimation

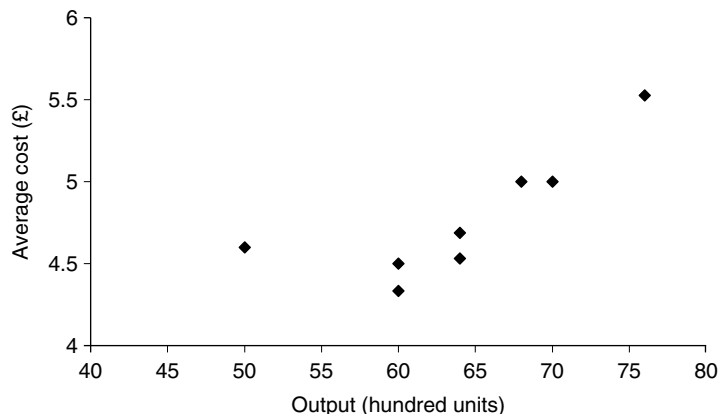
A study is carried out for DP Corp. regarding its costs and output over a period of eight months, during which time its capacity has remained constant. The results are:

|                      |    |    |    |    |    |    |    |    |
|----------------------|----|----|----|----|----|----|----|----|
| Output ('00)         | 50 | 64 | 60 | 64 | 70 | 60 | 68 | 76 |
| Total costs (£ '000) | 23 | 29 | 27 | 30 | 35 | 26 | 34 | 42 |

- Draw a graph of the average cost curve.
- Estimate an appropriate total cost function for the firm.
- Estimate total costs if the firm produces 8,000 units.
- What can be said about returns to scale in the industry? Give your reasoning.
- Estimate average costs at 8,000 units.
- Estimate the marginal cost of the 8,000th unit.

#### Solution

- Graph showing rising average costs.
- The linear total cost function is  $C = -16.95 + 0.7453Q$ ;  $R^2 = 92$  per cent  
The quadratic cost function is  $C = 69.16 - 2.027Q + 0.022Q^2$   
The **quadratic form** is more appropriate since it can be seen from the graph that average cost is increasing with output. A cubic function could also be fitted, if a computer were used to perform the necessary



calculations. This could then be compared with the quadratic function by examining the adjusted  $R^2$ .

- c.  $TC = \text{£}48,000$
- d. Nothing; the scale is not changing
- e.  $AC = \text{£}47,860/8,000 = \text{£}5.98$
- f.  $MC = -2.027 + 2(0.022)(80) = \text{£}1.49$

### SP7.3 Long-run cost estimation

A survey is carried out regarding the costs and outputs for eight firms in an industry that have different plant sizes. Some of the firms use a newer Hiflow (H) technology, whereas the others use the older Loflow (L) technology. The results are:

| Firm                 | A  | B   | C  | D   | E  | F   | G   | H   |
|----------------------|----|-----|----|-----|----|-----|-----|-----|
| Technology           | L  | L   | L  | H   | L  | H   | H   | H   |
| Output ('00)         | 60 | 134 | 90 | 178 | 84 | 200 | 152 | 108 |
| Total costs (£ '000) | 28 | 56  | 40 | 51  | 38 | 52  | 45  | 35  |

- a. Estimate an appropriate cost function in the industry.
- b. Does the newer technology significantly affect costs? If so, by how much?
- c. What can be said about returns to scale in the industry? Give your reasoning.
- d. Estimate costs for a firm using the newer technology that produces 12,000 units.
- e. Estimate the marginal cost of the 12,000th unit for the firm above.

*Solution*

- a. Specify power function for long run, and estimate two models: one ignoring technology; the other incorporating the differences in technology using a dummy variable ( $T = 1$  for Hiflow).

$$1 \ C = 3.995^{0.4946} \text{ or } \ln C = 1.385 + 0.4946 \ln Q; \text{ adj. } R^2 = 71.9 \text{ per cent}$$

$$2 \ \ln C = 0.147 + 0.787 \ln Q - 0.305T; \text{ adj. } R^2 = 97.6 \text{ per cent}$$

- b. **Yes**,  $t$ -statistic  $= -8.006$ ,  $p = 0.000$ .

Cost is reduced by 30.5 per cent, other things being equal.

- c. **Economies of scale**, since coefficient of  $\ln Q < 1$  every 1 per cent increase in output only increases total costs by 0.787 per cent.

- d.  $\ln C = 0.147 + 0.787 \ln(120) - 0.305 = 3.610$

$$C = \text{£}37,000$$

- e.  $MC = 0.4946(3.995)(120)^{(0.4946-1)} \times 1000/100 = \text{£}1.76$

## Summary

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- 1 Cost estimation is important for managerial decision-making for many reasons: determining the optimal marketing mix, determining the optimal output and scale of the firm, and various types of forecast which can be used for determining whether to accept contract offers or engage in mergers and acquisitions.
- 2 There are three main cost scenarios: the short run, the long run and the learning curve.
- 3 Statistical methods are the most sophisticated and popular methods for estimating costs.
- 4 Statistical methods involve essentially the same procedural steps as with demand studies.
- 5 Time-series studies are generally used more in short-run situations while cross-sectional studies are more common for long-run situations.
- 6 Many of the problems in the statistical estimation of cost functions are concerned with definition and measurement relating to the use of accounting data.
- 7 Polynomial cost functions are used more to specify short-run functions while power functions are used more to specify long-run ones. However, the quadratic form can be used for long-run average cost functions if there are economies of scale and diseconomies of scale in different ranges of output.
- 8 Empirical studies indicate that short-run total cost functions are often linear, with constant marginal costs; most economists interpret this as meaning that these firms are operating at substantially less than full capacity and are therefore not suffering from the law of diminishing returns.
- 9 Empirical studies indicate that the type of long-run cost function depends on the industry in which the firm is operating. Some industries, particularly those involved in mass production, have a greater incidence of technical economies of scale.
- 10 The learning curve is also more relevant in industries where mass production is used, where there is a significant amount of skilled labour, and where there are substantial interdependencies between supply and demand.

## Review questions

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- 1 Explain the relative advantages and disadvantages of time-series and cross-section studies for estimating cost functions in the short run.
- 2 Explain the relative advantages and disadvantages of time-series and cross-section studies for estimating cost functions in the long run.
- 3 Explain the relative advantages and disadvantages of using a polynomial cost function compared with a power function for a long-run analysis.

- 4 If a polynomial cost function is used for short-run analysis, how should one determine what degree of polynomial to use?
- 5 What would it mean if cumulative average cost were increasing over time?
- 6 What does an 80 per cent learning rate mean? If the learning rate is 70 per cent, is the firm (or individual) learning faster or slower?

## Problems

- 7.1 Data are collected from eight firms regarding their costs in the last year, each firm having the same level of fixed assets:
  - a. Estimate an appropriate cost relationship.
  - b. A firm of the same type is aiming to produce 48,000 units next year; estimate its total costs, stating any assumptions that you need to make.

|                          |     |     |     |     |     |     |     |     |
|--------------------------|-----|-----|-----|-----|-----|-----|-----|-----|
| Total costs (\$ million) | 245 | 260 | 272 | 265 | 248 | 282 | 295 | 288 |
| Output ('000)            | 31  | 35  | 38  | 36  | 32  | 40  | 44  | 42  |

- c. Estimate average costs for the firm next year.
  - d. What can be said about returns to scale for the firms in this industry?
  - e. Estimate the marginal cost of the last unit produced by the firm next year.
  - f. If the firm is currently charging \$8,000, estimate the break-even level of output for the firm.
  - g. Estimate the effect on costs for each firm in the industry of an increase in output of 5,000 units.
- 7.2 A survey of costs and output is carried out for eight firms in an industry that have different plant sizes. The results are:

|                |    |     |    |     |    |     |     |     |
|----------------|----|-----|----|-----|----|-----|-----|-----|
| Output (x100)  | 50 | 134 | 90 | 178 | 84 | 200 | 152 | 108 |
| Costs (\$'000) | 23 | 56  | 42 | 75  | 38 | 84  | 64  | 48  |

- a. Estimate an appropriate cost function in the industry.
  - b. Estimate costs for a firm producing 12,000 units.
  - c. What can be said about returns to scale in the industry? Give your reasoning.
  - d. Estimate average costs at 12,000 units.
  - e. Estimate the marginal cost of the 12,000th unit.
  - f. If the going price in the industry is \$4.60, calculate the break-even output.

7.3 The following cost models are estimated for a firm, based on short-run data:

|                                  | adj. $R^2$ |
|----------------------------------|------------|
| $C = -3,376.0 + 11.81Q$          | .886       |
| (9.320)                          |            |
| $C = 2,493 - 27.17Q + 0.126Q^2$  | .987       |
| (-6.115)      (8.815)            |            |
| $C = 1,595 - 8.461Q + 0.0003Q^3$ | .989       |
| (-3.920)      (9.556)            |            |

- a. Which is the best model to use? Explain your answer.
  - b. Interpret the coefficients in the model in terms of what they indicate.
  - c. What are the implications of your model in part (a) as far as break-even output is concerned?
- 7.4 MJ, a management consultant, is processing 500 questionnaires that the firm has received from respondents. MJ starts work on Monday at 9 a.m., takes an hour for lunch at 1p.m. and finishes at 5 p.m. The following progress is noted:

| Time  | Total questionnaires processed since 9 a.m. |
|-------|---------------------------------------------|
| 9.30  | 3                                           |
| 10.00 | 8                                           |
| 10.30 | 15                                          |
| 11.00 | 24                                          |

- a. Is there evidence of a learning curve? Give your reasons.
- b. Estimate and interpret the learning rate.
- c. How long did the first questionnaire take to process?
- d. Estimate the day and time when MJ will finish processing all the questionnaires.
- e. How long will the last questionnaire take to process?
- f. How many questionnaires will be processed by the end of Tuesday?

## Notes

- 1 G. Stigler, 'The economies of scale', *Journal of Law and Economics*, **1**(1958): 54-81.
- 2 J. Dean, *Statistical Cost Estimation*, Bloomington: Indiana University Press, 1976, pp. 3-35.
- 3 J. Johnston, *Statistical Cost Analysis*, New York: McGraw-Hill, 1960.
- 4 I. M. Dobbs, *Managerial Economics*, Oxford: Oxford University Press, 2000, p. 190.